1 Worksheet

Question 1.1 (Practice 1). Let $f \in L^2$ with $\sum_n |\hat{f}(n)| < \infty$. Show that there is a continuous function g such that $f = g$ a.e.

Proof. $\square$

Question 1.2 (Practice 2). Let $f : \mathbb{R} \to \mathbb{R}$ be 2π-periodic and in C^α. That is, $|f(x) - f(y)| \le C|x - y|^\alpha$ for some $C > 0$ and $\alpha \in (0, 1)$. Show that the Fourier series of f converges pointwise to f.

Proof. $\square$

Question 1.3 (Practice 3). Show that L^∞ is complete.

Proof. $\square$

Question 1.4 (Practice 4). Let $f : E \to \mathbb{R}$. If either $m(E) < \infty$ or $f \in L^p(E)$ for some $p \ge 1$, show that

$$\lim_{p\to\infty} \|f\|_{L^p(E)} = \|f\|_{L^\infty(E)}.$$

Find a counterexample if the assumptions are dropped.

Proof. $\square$

Question 1.5 (Practice 5). Let $p \in [1, \infty)$. Let A_p be the set of functions $f : [0, 1] \to \mathbb{R}$ that have a continuous derivative with

$$\int_0^1 |f'(x)|^p dx \le 1.$$

For which p are closed (with respect to the topology of uniform convergence) subsets of A_p compact?

Proof. $\square$

Question 1.6 (Practice 6). Let $p \in [1, \infty]$. Construct a function $f \in L^p(\mathbb{R})$ but $f \notin L^q(\mathbb{R})$ for any $q \neq p$.

Proof. $\square$

Question 1.7 (Practice 7). Let f_n be a sequence that converges in L^p to some f. Show that there exists a subsequence that converges a.e.

Proof. $\square$

Question 1.8 (Practice 8). This problem explores the concept of *convergence in measure*. Let $f_n : E \to \mathbb{R}$ be a sequence of measurable functions. We say that f_n converges to some f in measure if for all $\varepsilon > 0$ we have

$$\lim_{n\to\infty} m(\{x : |f_n(x) - f(x)| > \varepsilon\}) = 0.$$

(a) If f_n converges to f in L^p show that it converges in measure.

(b) If a sequence converges in measure, show that a subsequence converges pointwise a.e.

(c) If $m(E) < \infty$ and f_n converges pointwise, show that it converges in measure.

(d) Find sequences that converge in measure but not pointwise.

(e) Show that Fatou's Lemma and the MCT hold if pointwise convergence is replaced by convergence in measure.

Proof. □

Question 1.9 (Practice 9)**.** This problem shows that the Riemann integral coincides with the Lebesgue integral. In this problem let $f : [a, b] \to \mathbb{R}$ be a bounded function (not necessarily measurable).

(a) Let P_k be a sequence of partitions of $[a, b]$ whose mesh goes to 0 and are nested inside each other (i.e., P_{k+1} is a refinement of P_k for all k). For P_k let

$$G_k(x) = \sum_j M_j \mathbf{1}_{(t_{j-1}, t_j]}, \quad g_k(x) = \sum_j m_j \mathbf{1}_{(t_{j-1}, t_j]},$$

where the intervals $(t_{j-1}, t_j]$ are the partition of $[a, b]$ corresponding to P_k and M_j, m_j are the sup and inf of f over $[t_{j-1}, t_j]$. Show that the Lebesgue integrals of g_k and G_k correspond to the upper and lower Riemann sums associated to P_k and f.

(b) Let $G(x) = \lim_{k\to\infty} G_k(x)$ and $g(x) = \lim_{k\to\infty} g_k(x)$ (why do the limits exist?). Note $g \le f \le G$ everywhere. Show that if f is Riemann integrable, then $g = G$ a.e. (simply note that the Lebesgue integrals of G and g are equal to the limits of the Lebesgue integrals of G_k and g_k which coincide if f is Riemann integrable, as these integrals equal the upper and lower Riemann sums associated to the partitions P_k). Therefore, f is measurable as g and G are limits of measurable functions and its Lebesgue integral equals its Riemann integral.

(c) Define

$$H(x) = \lim_{\delta\to 0} \sup_{|y-x|\le\delta} f(y), \quad h(x) = \lim_{\delta\to 0} \inf_{|y-x|\le\delta} f(y).$$

Why do the above limits exist? Show that $H(x) = h(x)$ iff f is continuous at x.

(d) Show that the Lebesgue integrals of H and h are the limits of the upper and lower Riemann sums of f over any sequence of partitions whose mesh goes to 0. So if f is Riemann integrable, $H = h$ a.e. Therefore f is the a.e. limit of continuous functions and hence is Lebesgue measurable and its Lebesgue integral equals its Riemann integral.

Proof. □